Activity 11 Pre-Lab — Synthesis of Luminol (Forensic Chemistry)

Organic Lab · two-step synthesis + chemiluminescence

[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Synthesize **luminol** from 3-nitrophthalic acid in two steps, then test its **chemiluminescence** with a blood-substitute (iron catalyst) + hydrogen peroxide — the glow used in forensics to detect blood.

✍ **DRAW — the synthesis (2 steps)**

1) 3-nitrophthalic acid + hydrazine → **3-nitrophthalhydrazide** + 2 H_2O in triethylene glycol, heat to ~200 °C (distill off water)

2) 3-nitrophthalhydrazide + $\mathbf{Na_2S_2O_4}$ (in NaOH) → **luminol** (5-amino-2,3-dihydrophthalazine-1,4-dione) reduces $-NO_2 \rightarrow -NH_2$; then acidify w/ acetic acid to precipitate

CONTEXT ▸ Directions also ask you to **draw the mechanism** for the boxed portion of the reaction — leave room for it.

Chemical Reagents & Safety Table

CONTEXT ▸ Required (8): 3-nitrophthalic acid, hydrazine, triethylene glycol, aq. NaOH, sodium hydrosulfite, glacial acetic acid, potassium ferricyanide, 3% H_2O_2. Typical SDS values — cross-check the SDS tile. **Hydrazine is toxic/carcinogenic** — handle carefully.

Chemical	Role	MW	Amount	Physical properties	Hazards (GHS)
3-Nitrophthalic acid	Starting material	211.13	~1 g	Off-white / pale-yellow solid	H302 harmful if swallowed · H315 · H319 · H335
Hydrazine (8% aq)	Forms the hydrazide ring	32.05	2 mL	Colorless aqueous liquid	H301+H311+H331 toxic · H314 corrosive · H317 sensitizer · H350 may cause cancer · H410 aquatic
Triethylene glycol	High-boiling solvent	150.17	3 mL	Colorless viscous liquid; bp 285 °C	No significant GHS hazards (high-boiling solvent)
Sodium hydrosulfite (dithionite)	Reduces $-NO_2 \rightarrow -NH_2$	174.11	as directed	White powder	H251 self-heating (may ignite) · H302 · EUH031 toxic gas w/ acid
Sodium hydroxide (aq)	Base for reduction step	40.00	as directed	Colorless aqueous solution	H290 · H314 severe skin burns & eye damage
Glacial acetic acid	Acidify → precipitate luminol	60.05	as directed	Colorless liquid; bp 118 °C	H226 flammable · H314 severe skin burns & eye damage
Potassium ferricyanide	Catalyst (blood/iron mimic)	329.24	Part 2 (glow test)	Red crystalline solid	Low hazard · EUH032 — do NOT mix with acid (releases toxic gas)
Hydrogen peroxide (3%)	Oxidant for the glow	34.01	Part 2 (glow test)	Colorless liquid	H319 eye irritation (dilute)

GRAB FIRST: 50-mL round-bottom (no stir bar) · heating mantle + Variac + raised lab jack · simple-distillation head + **high-temp thermometer (≥200 °C)** + condenser + receiver · boiling stones · Erlenmeyer + steam bath (heat 15 mL water) · ice bath · Büchner + filter flask + vacuum · 3-mL syringe (hydrazine) + 25-mL Erlenmeyer.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Reaction 1 — nitro-luminol

1. Heat **15 mL water** on a steam bath (boiling stones) — needed later (step 5).
2. Put **~1 g 3-nitrophthalic acid** in a **50-mL RBF** (no stir bar); clamp; lower into a heating mantle on a raised lab jack.
3. At the hydrazine hood (nitrile gloves) draw **2 mL 8% hydrazine** into a syringe; add it to the flask. Hydrazine is toxic + a carcinogen — gloves, no skin/eye contact, fresh gloves after.
4. Add a few boiling stones; gently heat (Variac 75–80%) until the acid dissolves. Add **3 mL triethylene glycol**.
5. Build a **simple distillation** (high-temp thermometer **submerged** in the solution, not touching bottom). Instructor checks → Variac 80%, heat until the vapor/solution reads **200 °C** (water distills off) = reaction done. Don't adjust thermometer while hot — remove, tweak, replace.
6. Remove from heat, cool to ~100 °C, add the **15 mL hot water**; cool in an ice bath; **vacuum-filter** the light-yellow nitro compound (no need to dry).

Reaction 2 — reduce to luminol

7. Dissolve the nitro compound in **NaOH**, add **sodium hydrosulfite** (heat as directed) to reduce $-NO_2 \rightarrow -NH_2$; then **acidify with acetic acid** to precipitate **luminol**; vacuum-filter, weigh → % yield.

Part 2 — chemiluminescence

8. Make the luminol glow: basic luminol + **3% H_2O_2** + **$K_3Fe(CN)_6$** catalyst → blue light. Observe/record. Keep $K_3Fe(CN)_6$ away from acid.

✍ **LEAVE ROOM**
for the required mechanism drawing of the boxed reaction step.

Experiment Notes (in lab)

Mass 3-nitrophthalic acid: ________ g

Rxn 1 — max temp reached / time: ________

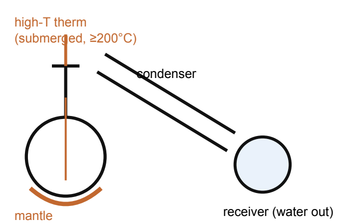
Nitro compound appearance:

Rxn 2 — color changes on reduction/acidify:

Luminol mass / % yield: ________

Glow test — color/brightness/duration:

Setup — draw these



simple distillation (Rxn 1)

CONTEXT (don't copy) ›

Why distill to 200 °C? the hydrazide ring only forms at high temp; triethylene glycol (bp 285 °C) lets you get there, and boiling off the water pushes the equilibrium to product.

Why does luminol glow? H_2O_2 (with the iron catalyst) oxidizes luminol to an excited 3-aminophthalate that drops to ground state and emits blue light — the same reaction forensics uses to find blood (iron in hemoglobin is the catalyst).

Clean-up & Conclusions

- Hydrazine rinses → labeled hydrazine waste; distillate water → aqueous waste; keep ferricyanide from acid; organic/aqueous to correct waste; bench task.
- **Conclusions** (~4 sentences after lab): overall % yield of luminol; what each step did (ring formation, then $-NO_2 \rightarrow -NH_2$ reduction); what the glow test demonstrated (H_2O_2 + iron catalyst → chemiluminescence); a loss/error source.